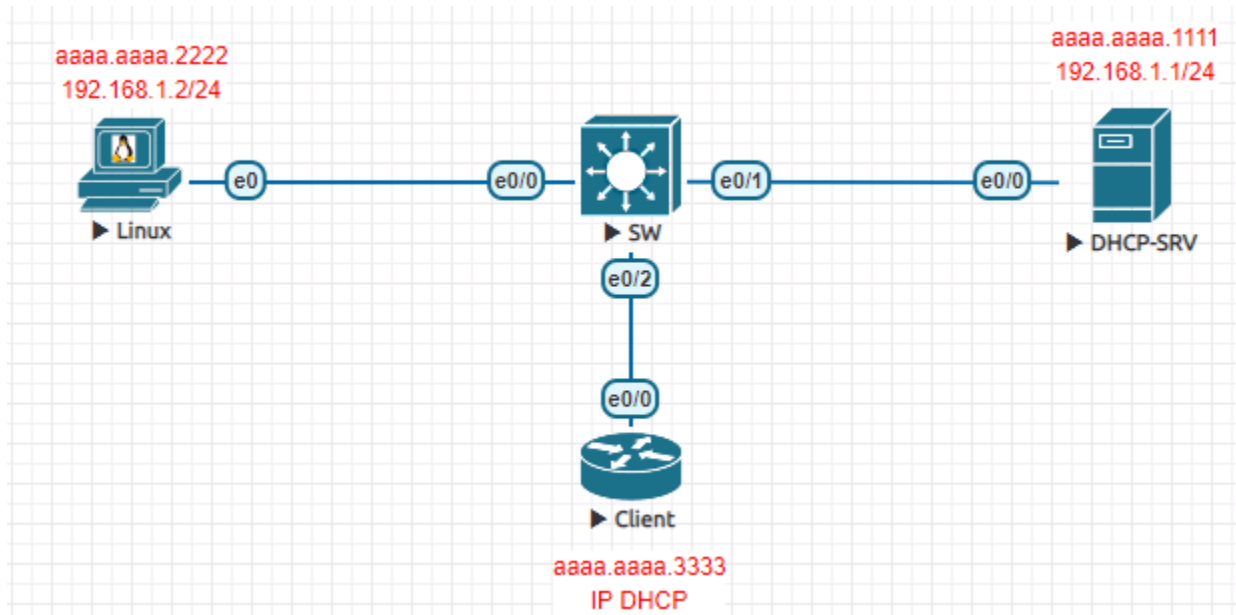


## ARP Poisoning or Spoofing Attack:



Lets start Ettercap graphically type below command in Kali Linux Terminal.

```
root@kali: ~  
root@kali: ~#  
root@kali: ~# ettercap -G
```

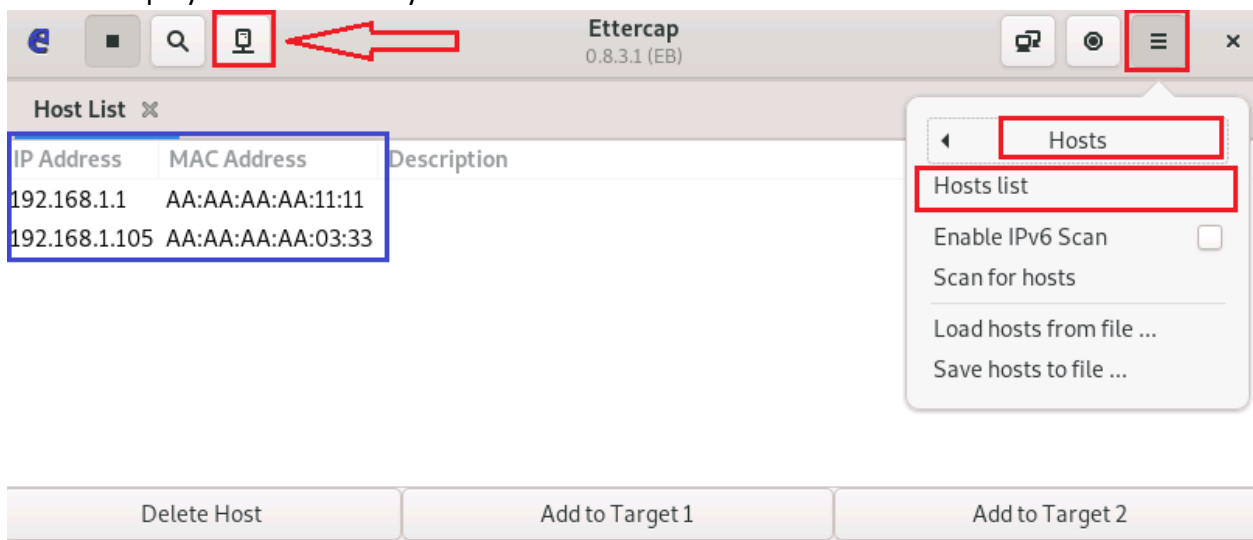
Now, click on tick mark to start Unified Sniffing on Kali Linux.



Click to Scan for Hosts in the Network you can use shortcut menu as well.



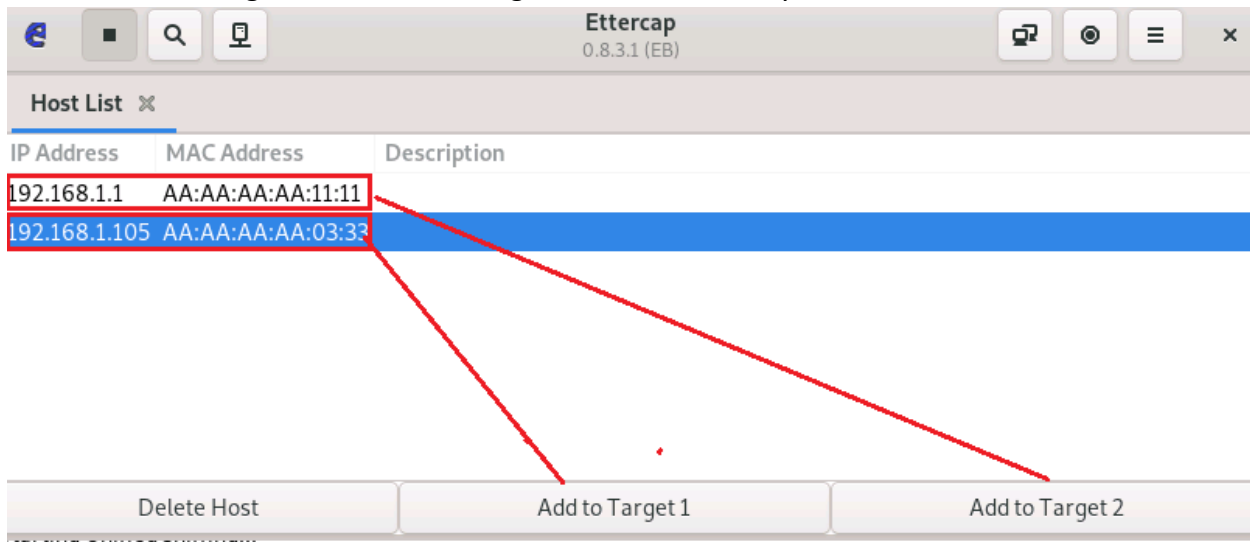
Click to display Scan Host Lists you can use shortcut button on menu as well.



.ua: no scripts were specified, not starting up!  
Starting Unified sniffing...

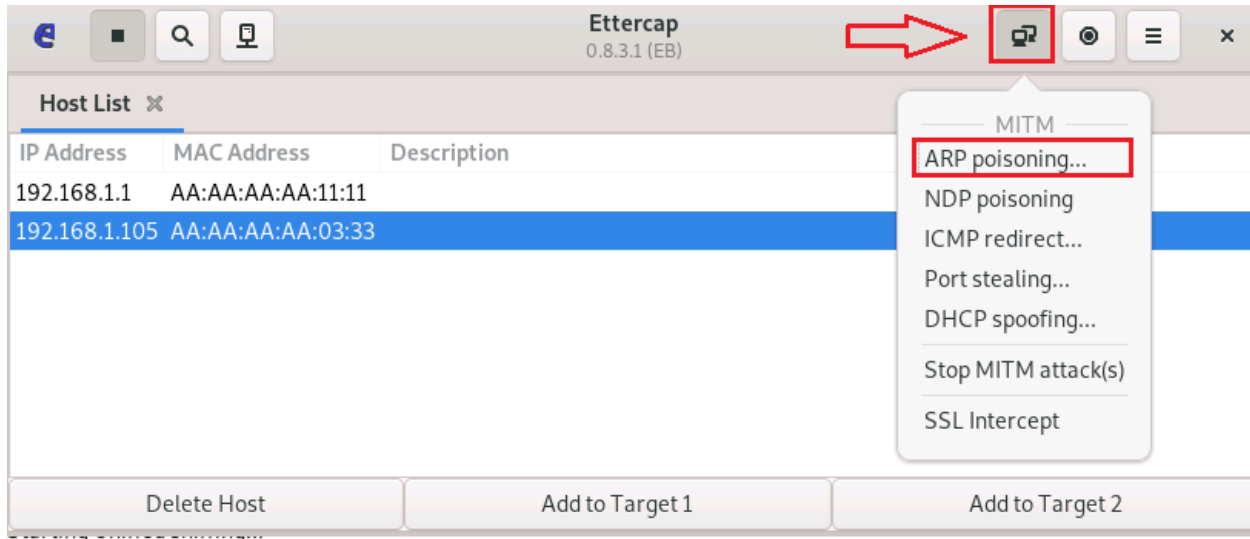
Randomizing 255 hosts for scanning...  
Scanning the whole netmask for 255 hosts...  
2 hosts added to the hosts list...

Click to Add to Target 1 and Add to Target 2 to make it ready for MITM attack.



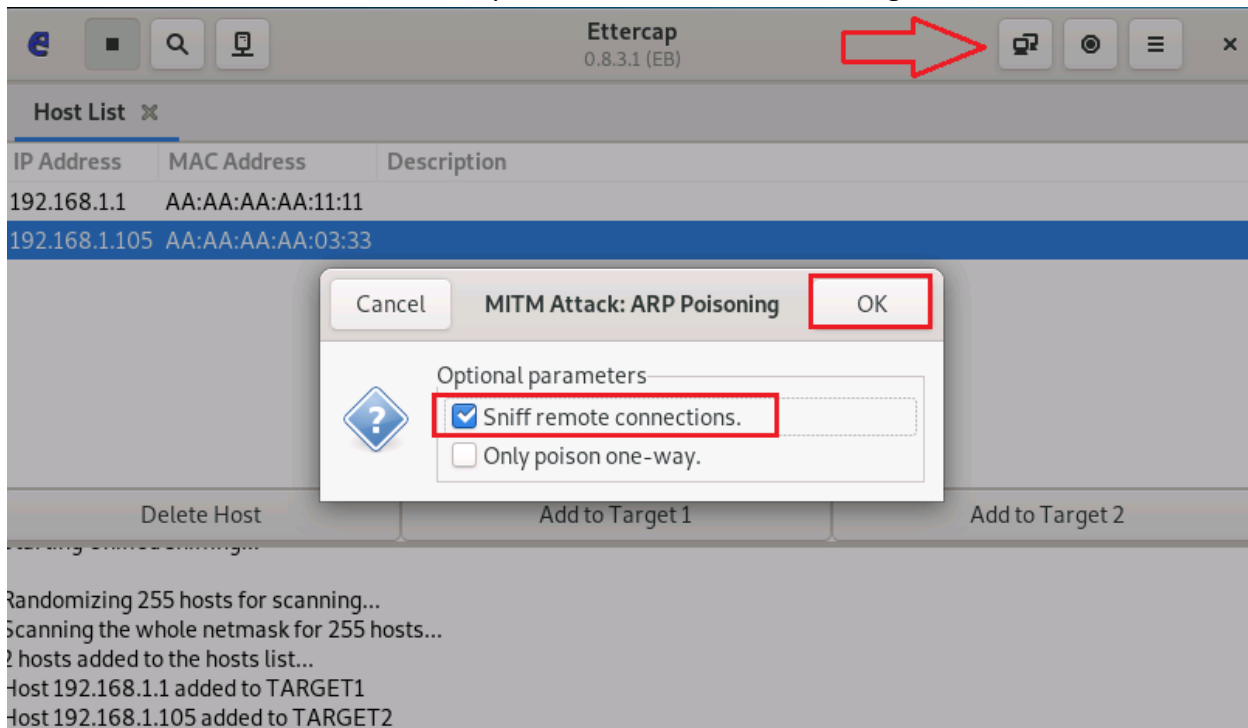
Randomizing 255 hosts for scanning...  
Scanning the whole netmask for 255 hosts...  
2 hosts added to the hosts list...  
Host 192.168.1.1 added to TARGET1  
Host 192.168.1.105 added to TARGET2

Click to start ARP Poisoning attack on the give Target Systems.

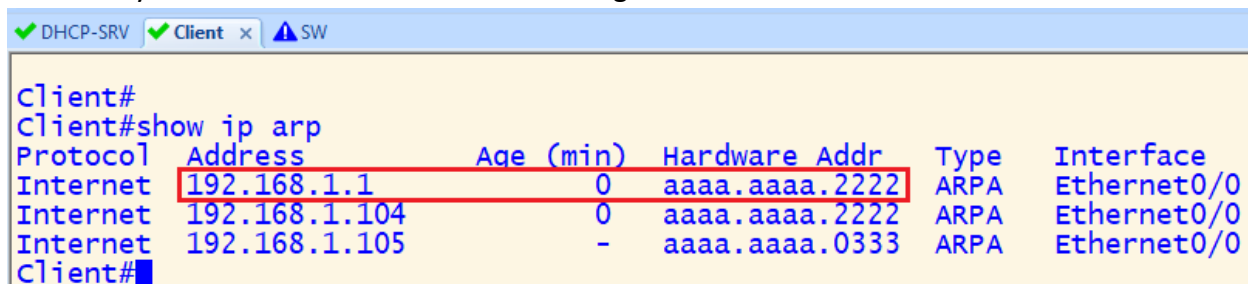


Randomizing 255 hosts for scanning...  
Scanning the whole netmask for 255 hosts...  
2 hosts added to the hosts list...  
Host 192.168.1.1 added to TARGET1  
Host 192.168.1.105 added to TARGET2

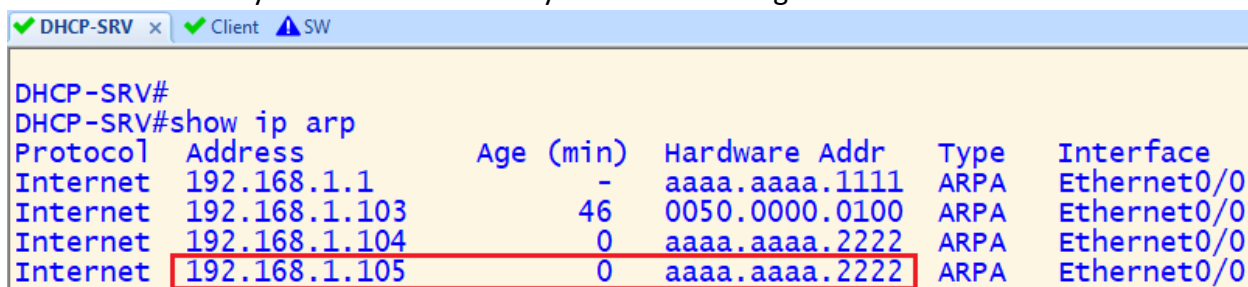
Select Sniff Remote Connections and press OK to start ARP Poisoning Attack.



In Client system front of DHCP-SRV it is showing Kali Linux MAC Address.



And In DHCP-SRV system front of Client System it is showing Kali Linux MAC Address.



2<sup>nd</sup> Method for ARP Spoofing Attack first command is Kali Linux start work as a router.

```
root@kali:~# echo 1 > /proc/sys/net/ipv4/ip_forward
root@kali:~# arpspoof -i eth0 -t 192.168.1.105 -r 192.168.1.1
```

### Dynamic ARP Inspection:

- o DAI stands for Dynamic Address Resolution Protocol Inspection.
- o DAI is a security feature that rejects invalid and malicious ARP packets.
- o DAI feature prevent man-in-the-middle attacks such as ARP poisoning.
- o DIA feature prevent man-in-the-middle attack such as ARP Spoofing.
- o DAI relies on Dynamic Host Control Protocol (DHCP) snooping feature.
- o DHCP snooping builds binding database of valid MAC address, IP & VLAN interface.
- o DAI uses the DHCP snooping binding database to validate bindings.
- o Dynamic ARP Inspection (DAI) verifies IPv4 address to MAC address bindings.
- o If mismatch happened on untrusted port, DAI will discard spoofed ARP packets.
- o Dynamic ARP Inspection (DAI) only inspects ARP packets from untrusted ports.
- o Dynamic ARP Inspection (DAI) can be enabled globally per VLAN.

Before Apply Dynamic ARP Inspection (DAI) it require below Configuration.

|                                                                                                                                                                                         |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DHCP SERVER Configuration</b>                                                                                                                                                        |
| DHCP-SRV(config)#interface ethernet0/0<br>DHCP-SRV(config-if)#ip address 192.168.1.1 255.255.255.0<br>DHCP-SRV(config-if)#mac-address aaaa.aaaa.1111<br>DHCP-SRV(config-if)#no shutdown |
| DHCP-SRV (config)#ip dhcp excluded-address 192.168.1.1 192.168.1.100<br>DHCP-SRV (config)#ip dhcp pool TEST<br>DHCP-SRV (dhcp-config)#network 192.168.1.0 /24                           |
| DHCP-SRV # show ip dhcp binding                                                                                                                                                         |

|                                                                                                                        |
|------------------------------------------------------------------------------------------------------------------------|
| <b>DHCP Client Configuration</b>                                                                                       |
| Router(config)#hostname client<br>client(config)#interface ethernet0/0<br>client(config-if)#mac-address aaaa.aaaa.3333 |
| client(config)#interface ethernet0/0<br>Client(config-if)#no ip address<br>Client(config-if)#ip address dhcp           |
| Client# show ip int br                                                                                                 |

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| <b>Kali Linux Configuration</b>                                                                              |
| root@kali:~# ifconfig eth0 192.168.1.2 netmask 255.255.255.0<br>root@kali:~# macchanger -m aa:aa:aa:aa:22:22 |

|                                                                       |
|-----------------------------------------------------------------------|
| <b>DHCP Snooping Configuration SW</b>                                 |
| SW(config)#ip dhcp snooping                                           |
| SW(config)#ip dhcp snooping vlan 1                                    |
| SW(config)#interface Ethernet 0/1                                     |
| SW(config-if)#ip dhcp snooping trust                                  |
| SW(config-if)# no ip dhcp snooping information option                 |
| SW(config)#interface Ethernet 0/1                                     |
| SW(config-if)# ip dhcp snooping limit rate 100                        |
| SW(config)#interface range Ethernet 0/0, Ethernet 0/2                 |
| SW(config-if)# ip dhcp snooping limit rate 20 (20 packets per second) |
| SW# show ip dhcp snooping binding                                     |
| SW# show ip dhcp snooping                                             |
| SW# show ip dhcp snooping statistics                                  |

|                                                 |
|-------------------------------------------------|
| <b>Dynamic ARP Inspection Configuration SW1</b> |
| SW(config)#ip arp inspection vlan 1             |
| SW(config)# interface Ethernet 0/1              |
| SW(config-if)# ip arp inspection trust          |
| SW# show ip arp inspection                      |

|                                                          |
|----------------------------------------------------------|
| <b>Change Client IP Address</b>                          |
| Client(config)#interface e0/0                            |
| Client(config-if)#ip address 192.168.1.103 255.255.255.0 |
| Client# ping 192.168.1.2                                 |
| Client(config)#interface e0/0                            |
| Client(config-if)#ip address dhcp                        |

SW#**show ip arp inspection**

|                            |               |              |               |                     |
|----------------------------|---------------|--------------|---------------|---------------------|
| Source Mac Validation      | : Disabled    |              |               |                     |
| Destination Mac Validation | : Disabled    |              |               |                     |
| IP Address Validation      | : Disabled    |              |               |                     |
| Vlan                       | Configuration | Operation    | ACL Match     | Static ACL          |
| ---                        | -----         | -----        | -----         | -----               |
| 1                          | Enabled       | Active       |               |                     |
| Vlan                       | ACL Logging   | DHCP Logging | Probe Logging |                     |
| ---                        | -----         | -----        | -----         |                     |
| 1                          | Deny          | Deny         | off           |                     |
| Vlan                       | Forwarded     | Dropped      | DHCP Drops    | ACL Drops           |
| ---                        | -----         | -----        | -----         | -----               |
| 1                          | 11            | 263          | 263           | 0                   |
| Vlan                       | DHCP Permits  | ACL Permits  | Probe Permits | Source MAC Failures |
| ---                        | -----         | -----        | -----         | -----               |
| 1                          | 7             | 0            | 2             | 0                   |